APPLICATION PORTFOLIO

Sound Design Portfolio

Hu JinLin

Recording Arts undergraduate
Xinghai Conservatory of Music · 2023-2027 (expected)

Application to the University of Glasgow
Creative Sound Design & Audiovisual Media MSc

Three selected works in interactive audio and sound for picture. The accompanying website presents full recordings first, followed by audio implementation demonstrations and source notes on each project page.

Selected works

01 **Hua Shou** pp. 2-3
Pencil gestures and paper-textured combat.

02 **Wake Up Tokyo** pp. 4-5
Wwise events and a Steam Audio footstep comparison.

03 **Delta Force PV** p. 6
Independent music and sound-effects redesign.

Supplementary project: **ForgeaX Sound Workshop** (p. 7).
Roles, source credits and listening links are collected on p. 8.

Open the portfolio website

01 / HUA SHOU

Drawing as an audio control

Team game · Unity and Wwise · 2026

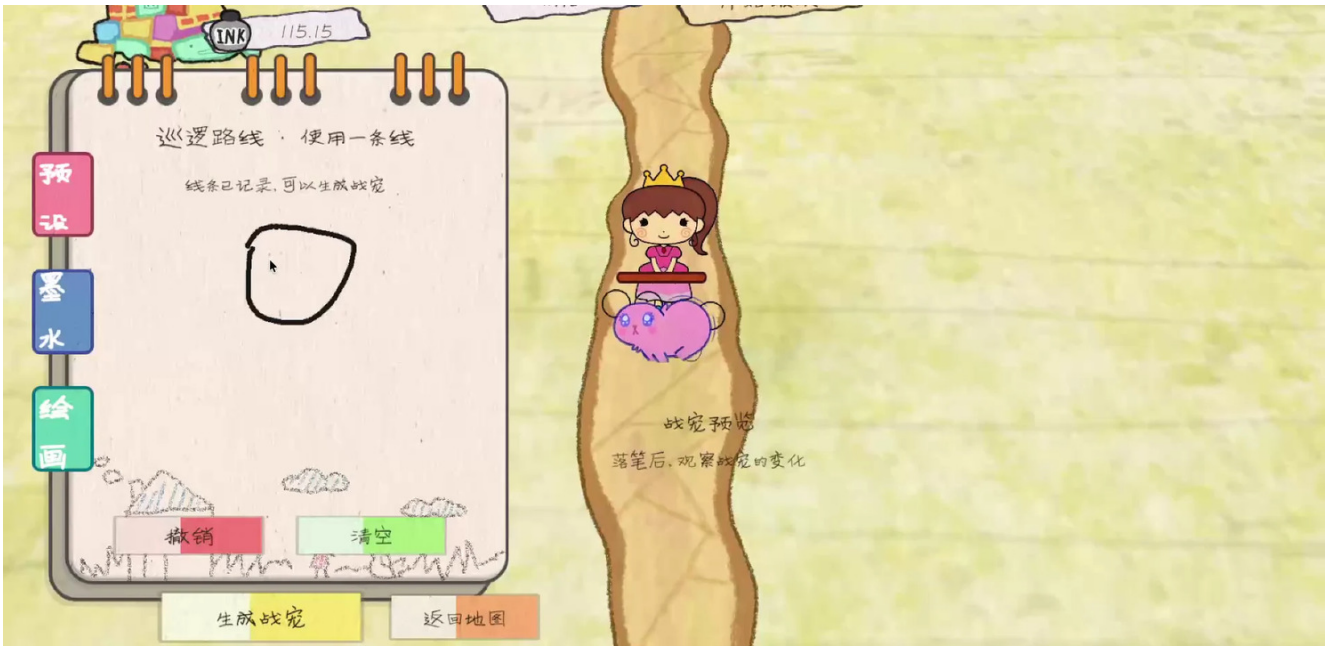


Figure 1. The player draws a patrol route before summoning a creature. Gameplay recording, 00:25.

My contribution

I selected and edited the pencil and combat sounds, implemented drawing controls in Unity and Wwise, and balanced the cues against music and combat feedback.

Sound decisions

Drawing. One friction loop follows a moving stroke. Direction changes trigger short accents; pausing the hand stops the loop. This keeps a long line continuous while making a bend audible.

Combat. A sweep marks the pink creature's strike. I moved a paper-tear layer into the release so the material texture follows the impact rather than competing with its attack.

Wave flow. Preparation and combat call separate music cues. Drawing and danger feedback take priority over repeated action sounds.

Watch gameplay · 02:24

Watch the Wwise demonstration

01 / HUA SHOU · IMPLEMENTATION

Loop, turn and release

Implemented in the playable version: a friction loop and turn accents.

Gesture mapping

Pen_Speed	Drawing speed controls the friction layer's level.
Pen_Turn	A turn briefly lowers the loop beneath an accent.
Turn event	One of four variants; minimum retrigger interval 240 ms.
Stop conditions	No movement for about 45 ms, release, pause or focus loss.

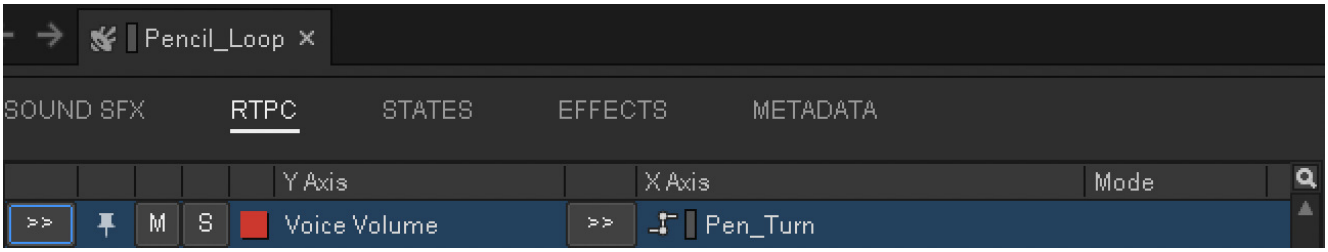


Figure 2. Wwise Authoring: Pencil_Loop binds Pen_Turn to Voice Volume.

A brief dip under the turn accent

On a full-strength turn, the requested Pen_Turn reaches 100 at 90 ms, holds until 160 ms and returns to zero by 320 ms. The Wwise curve maps 0 / 50 / 100 to 0 / -5 / -12 dB on the loop; its parameter smoothing is 20 ms.

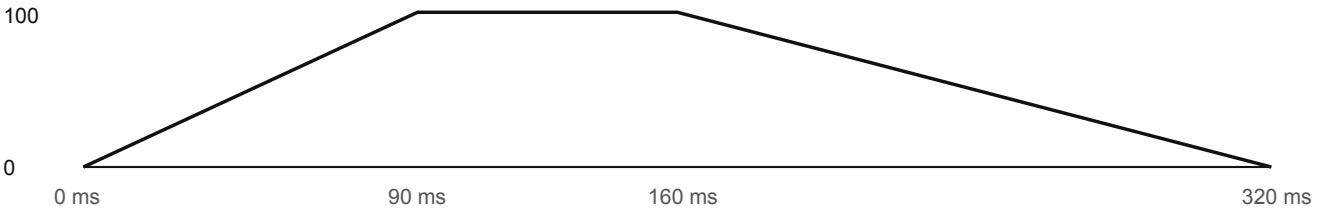


Figure 3. Requested Pen_Turn envelope, before Wwise smoothing. The vertical axis is the RTPC value.

Paper-tail edit

I cut 0.17-0.53 s from the Office Building paper-tear recording. The two attack variants place this 360 ms slice at 250 ms and 290 ms, with a 20 ms fade-in, 70 ms fade-out and an 8 dB layer reduction.

Inspect the implementation and source notes

02 / WAKE UP TOKYO

Feedback in a growing crowd

Team game · Browser, Wwise Web and Steam Audio · 2026



Figure 4. Street gameplay. The opening portfolio video is my continuous 3:41 recording with game audio.

My contribution

I edited and implemented the melee, progression and movement cues, prepared the music, and integrated Wwise and Steam Audio into the browser build. Game programming and visuals are team work.

Wwise events and music

Light and heavy punches use separate events. Card reveal and upgrade confirmation provide short feedback during progression. The edited demonstration isolates these cues in Wwise from 00:39, then returns to gameplay.

Music_Street starts the street cue with a 0.70 s fade-in and stops seven other phases with 0.65 s fades. Radio-triggered, frequency-selective ducking creates space while the music continues. The soundtrack includes selected and edited Suno-generated material.

Watch my continuous gameplay · 03:41

Watch audio implementation · 02:36

Play Tokyo in the browser

02 / WAKE UP TOKYO · STEAM AUDIO

A controlled moving-source comparison

Direct-sound HRTF · Wwise 2025.1 · Steam Audio 4.8.1

Steam Audio’s HRTF renderer filters each ear according to source direction. I used footsteps to compare it with Wwise panning while keeping the event selection and distance attenuation in Wwise [1, 2].

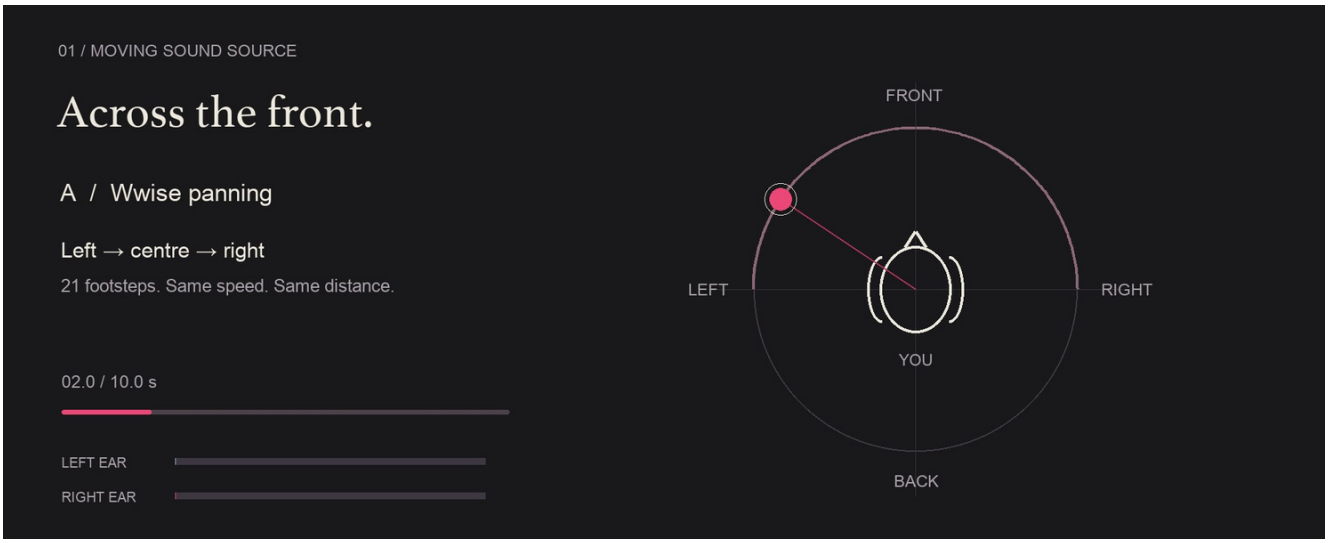


Figure 5. The moving-source demonstration. A pass across the front is followed by a pass behind the listener.

Comparison method

Each pass uses 21 road footsteps on the same three-metre-radius path, at the same speed, with a fixed listener and repeatable event sequence. One constant gain per renderer matches the overall listening level. The audio was rendered offline through the project’s actual runtime.

One RTPC, two routes

0 · Steam Audio	Spatializer active; native Wwise panning blend 0%.
100 · Wwise	Spatializer bypassed; native panning blend 100%.

With headphones, compare front-back placement and the timbral change through the centre. This demonstrates direct-sound rendering, not measured localisation accuracy. Reflections, occlusion and mobile performance are outside this evaluation.

Listen to the moving A/B · 00:46

Inspect the Wwise routing

03 / DELTA FORCE PV

Sound for a fixed picture edit

Independent redesign · Music editing, sound effects and final mix · 2026



Figure 6. Still from the source cinematic. The original picture and edit belong to their respective rights holders.

Scope and authorship

I used a publicly released Delta Force season cinematic as the fixed picture for an independent sound redesign. My work covers music and sound-effects editing, sound-to-picture synchronisation and the final mix. I did not make the original film or its visual edit.

Listening focus

Contact. Follow the attacks of mechanical details and impacts against their visible actions. Their tails carry the sound beyond the point of contact.

Mix balance. Listen for changes between the music and close action sounds. The score spans the sequence, while individual effects move forward at the corresponding visual events.

Watch the complete sound redesign · 02:52

Open the project page

SUPPLEMENTARY / FORGEAX SOUND WORKSHOP

An audio review workflow

Team project · Audio workflow design and implementation with collaborators



Figure 7. Sound Workshop production interface. Product scope is cross-referenced with the official ForgeaX catalogue [3].

My contribution

I led the audio workflow design and worked with developer collaborators and AI-assisted coding on implementation. The workflow connects a scene brief, candidate audio, revision and game-event playback.

From a candidate to a game event

Music, effects and voice share a review step. The creator can audition a take, adjust it and revisit its source and settings. A chosen version is then connected to a playback trigger identified in the game code.

One documented test followed a 16-second generated music cue through export and import into a game template, then checked triggering, looping and stopping. This demonstrates the delivery workflow; it is presented separately from the three finished sound examples.

Open the Sound Workshop case study

Official ForgeaX plugin catalogue

CREDITS AND ACCESS

Contributions, sources and links

Hua Shou

Game design and visual art: Hua Shou team. Sound design, editing and interactive audio: Hu JinLin. Pencil and paper textures: Office Building via Splice; sweep layers: Fantasy Game 2. Music includes Suno-generated material selected, edited and integrated by me.

Wake Up Tokyo

Game and visuals: Wake Up Tokyo team. Audio editing, Wwise authoring and browser audio integration: Hu JinLin, with AI-assisted implementation. Music includes Suno-generated material. Gunfire design is not presented as my work. Steam Audio is developed by Valve.

Delta Force PV

Independent sound redesign by Hu JinLin. Original picture: Delta Force season promotional cinematic, credited to the original rights holders. The portfolio presents my replacement music and sound-effects track.

Sound Workshop

Audio workflow and product direction: Hu JinLin. Implementation with developer collaborators and AI-assisted coding. Generated audio is identified as such. Official product description and icon: ForgeaX.

References

[1] Valve. Steam Audio: Spatializer.

[2] Valve. Steam Audio C API: Programmer’s Guide.

[3] ForgeaX. Sound Workshop, plugin catalogue.

Technical references accessed 9 November 2026. Project parameters and figures come from the Wwise authoring data, edit records and recordings used in these case studies.

Playback and game downloads

The website places implementation notes below each game’s introduction and recording. Tokyo is playable in a desktop browser. Download links for both games are on their project pages. Headphones are recommended for the moving-source comparison.

Portfolio and listening index